

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO1 – Imitation (L1)	Assessment:	Assessment Tool 1 – Pattern Recognition and Text Processing
Weightage:	40% of CIA		
CO5 Statement:	Analyse regular expression patterns. (BL-3)		
Student Name:		Reg. No.:	
Section / Batch:		Date:	

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- Show all intermediate steps, calculations, state transitions, and reasoning wherever applicable.
- Use only the Python re (Regular Expressions) module to solve the given problems.
- Clearly mention the Regular Expression (Regex) pattern used before writing the corresponding Python program.
- Display the required output for each question, including extracted values, validation results, counts, and positions wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
Regular Expression Pattern Generation	Pattern Matching Extraction Information	1
Search for a String	Keyword Search and Text Analysis	2
Split the documentation into sentences and words	Text Tokenization and Sentence Segmentation	3
Email and Strong Password Validation	Data Validation and Format Verification	4
Compile Once, Validate Many	Pattern Reusability and Performance Optimization	5

Section 1: Regular Expression Pattern Generation

Student Details:

Name: John Doe

Email: john.doe123@gmail.com

Mobile: +91-9876543210

Password: P@ssw0rd123

Date of Birth: 15/08/2004

Register Number: 23AIML1056

Department: Artificial Intelligence and Machine Learning

For the given student details formulate Regex patterns for the following:

Email ID

Mobile Number

Strong Password

Date of Birth (DD/MM/YYYY)

Register Number

Student Details

- **Name:** John Doe
- **Email:** john.doe123@gmail.com
- **Mobile:** +91-9876543210
- **Password:** P@ssw0rd123
- **Date of Birth:** 15/08/2004
- **Register Number:** 23AIML1056
- **Department:** Artificial Intelligence and Machine Learning

Regular Expression Patterns

1. Email ID Regex Pattern

Regex Pattern:

`^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$`

Description:

Validates an email address containing a username, '@' symbol, domain name, and domain extension.

2. Mobile Number Regex Pattern

Regex Pattern:

`^\+91-[6-9]\d{9}$`

Description:

Validates an Indian mobile number with country code (+91) followed by a 10-digit mobile number starting with 6, 7, 8, or 9.

3. Strong Password Regex Pattern

Regex Pattern:

`^(?=.*[a-z])(?=.*[A-Z])(?=.*\d)(?=.*[@#%^\&+=!]).{8,}$`

Description:

Validates a strong password containing at least one uppercase letter, one lowercase letter, one digit, one special character, and a minimum length of 8 characters.

4. Date of Birth (DD/MM/YYYY) Regex Pattern

Regex Pattern:

`^(0[1-9]|[12][0-9]|3[01])/(0[1-9]|1[0-2])/(19|20)\d{2}$`

Description:

Validates a date in the DD/MM/YYYY format.

5. Register Number Regex Pattern

Regex Pattern:

`^d{2}[A-Z]{4}d{4}$`

Description:

Validates a register number consisting of 2 digits, followed by 4 uppercase letters, and ending with 4 digits.

Validation of Student Details

Field	Value	Status
Email ID	john.doe123@gmail.com	Valid
Mobile Number	+91-9876543210	Valid
Password	P@ssw0rd123	Valid
Date of Birth	15/08/2004	Valid
Register Number	23AIML1056	Valid

Section 2: Search for a String

Artificial Intelligence (AI) is transforming industries across the world. AI is used in healthcare to assist doctors in diagnosis, in banking to detect fraud, and in education to provide personalized learning experiences. Many companies invest heavily in AI research because AI improves efficiency and enables intelligent decision-making. As AI continues to evolve, professionals with AI skills are in high demand. Write a Python program using the re module to perform the following operations for the word "AI":

1. Find the first occurrence of the word "AI" using re.search().
2. Display the starting position of the first occurrence.
3. Display the ending position of the first occurrence.
4. Display the total number of times the word "AI" appears in the passage.
5. Print an appropriate message if the word is not found.

Problem Statement

Artificial Intelligence (AI) is revolutionizing various industries by enhancing efficiency, accuracy, and decision-making. In the healthcare sector, AI supports medical professionals in diagnosing diseases, while in banking it helps identify fraudulent transactions. In education, AI enables personalized learning by adapting to individual student needs. As organizations increasingly adopt AI-driven solutions and invest in research and development, the demand for skilled AI professionals continues to grow across multiple domains.

```
import re

text = """Artificial Intelligence (AI) is transforming industries across the world. AI is used in healthcare to assist doctors in diagnosis, in banking to detect fraud, and in education to provide personalized learning experiences. Many companies invest heavily in AI research because AI improves efficiency and enables intelligent decision-making. As AI continues to evolve, professionals with AI skills are in high demand."""

match = re.search(r"\bAI\b", text)

if match:
    print("First occurrence of 'AI' found.")
    print("Starting position:", match.start())

    print("Ending position:", match.end())

    occurrences = re.findall(r"\bAI\b", text)
    print("Total occurrences of 'AI':", len(occurrences))
else:
    print("The word 'AI' was not found in the passage.")
```

```
=== RESTART: C:/Users/Pooja Sri/AppData/Local/
First occurrence of 'AI' found.
Starting position: 25
Ending position: 27
Total occurrences of 'AI': 6
|
```

Section 3: Split the documentation into sentences and words

Refer the previous passage and Write a Python program using the re.split() method to perform the following operations:

- Split the passage into sentences using appropriate punctuation (., ?, !) as delimiters.
- Count and display the total number of sentences.

- Split the passage into words by considering one or more whitespace characters as delimiters.
- Count and display the total number of words.
- Display the list of extracted sentences and words.

Problem Statement

Artificial Intelligence (AI) has become a key technology driving innovation across numerous industries. It is widely applied in healthcare for disease diagnosis, in the banking sector for fraud detection, and in education to deliver customized learning experiences. Organizations are increasingly investing in AI technologies to improve productivity, automate complex tasks, and support data-driven decision-making. With the rapid advancement of AI, the need for professionals possessing AI expertise continues to rise across various sectors.

```
3.py - C:/Users/Pooja Sri/AppData/Local/Programs/Python/Python313/3.py (3.13.9)
File Edit Format Run Options Window Help
import re

text = """Artificial Intelligence (AI) is transforming industries across the world. AI is used in healthcare to assist doctors in diagnosis,
banking to detect fraud, and in education to provide personalized learning experiences. Many companies invest heavily in AI
research because AI improves efficiency and enables intelligent decision-making. As AI continues to evolve, professionals with AI
skills are in high demand."""

sentences = re.split(r'[.!?]+', text)
sentences = [s.strip() for s in sentences if s.strip()]

print("List of Sentences:")
for i, sentence in enumerate(sentences, 1):
    print(f"{i}. {sentence}")

print("\nTotal Number of Sentences:", len(sentences))

words = re.split(r'\s+', text.strip())

print("\nList of Words:")
print(words)

print("\nTotal Number of Words:", len(words))

>>>
=== RESTART: C:/Users/Pooja Sri/AppData/Local/Programs/Python/Python313/3.py ===
List of Sentences:
1. Artificial Intelligence (AI) is transforming industries across the world
2. AI is used in healthcare to assist doctors in diagnosis, in
banking to detect fraud, and in education to provide personalized learning experiences
3. Many companies invest heavily in AI
research because AI improves efficiency and enables intelligent decision-making
4. As AI continues to evolve, professionals with AI
skills are in high demand
Total Number of Sentences: 4
List of Words:
['Artificial', 'Intelligence', '(AI)', 'is', 'transforming', 'industries', 'across', 'the', 'world.', 'AI', 'is', 'used', 'in', 'healthcare', 'to', 'assist', 'doctors', 'in', 'diagno',
s', 'in', 'banking', 'to', 'detect', 'fraud,', 'and', 'in', 'education', 'to', 'provide', 'personalized', 'learning', 'experiences.', 'Many', 'companies', 'invest', 'heavily', 'in',
AI', 'research', 'because', 'AI', 'improves', 'efficiency', 'and', 'enables', 'intelligent', 'decision-making.', 'As', 'AI', 'continues', 'to', 'evolve,', 'professionals', 'with', 'I',
', 'skills', 'are', 'in', 'high', 'demand.']
Total Number of Words: 60
```

Section 4: Email and Strong Password Validation

A website requires users to register by providing an Email ID and a Strong Password. Write a Python program using the re module to validate the following inputs.

Validation Criteria:

Email ID

- Must begin with one or more letters or digits.
- May contain ., _, %, +, or - before the @ symbol.
- Must contain a valid domain name.
- Must end with a valid domain extension such as .com, .org, .edu, or .in.

Strong Password

- Must contain at least 8 characters.
- Must contain at least one uppercase letter.
- Must contain at least one lowercase letter.
- Must contain at least one digit.
- Must contain at least one special character from @, #, \$, %, &, !, or *.
- Must not contain any whitespace characters.

Problem Statement

A website requires users to register using an **Email ID** and a **Strong Password**. Write a Python program using the **re** module to validate the email and password based on the given rules. The program should check whether the entered Email ID and Password are **Valid** or **Invalid**.

4.py - C:/Users/Pooja Sri/AppData/Local/Programs/Python/Python313/4.py (3.13.9)

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```
import re
email = input("Enter Email ID: ")
password = input("Enter Password: ")
email_pattern = r'^[A-Za-z0-9]+[A-Za-z0-9._%+-]*@[A-Za-z0-9.-]+\.(com|org|edu|in)$'
password_pattern = r'^(?=.*[A-Z])(?=.*[a-z])(?=.*\d)(?=.*[@#%&!*])[A-Za-z\d@#%&!*]{8,}$'
if re.fullmatch(email_pattern, email):
    print("Valid Email ID")
else:
    print("Invalid Email ID")
if re.fullmatch(password_pattern, password):
    print("Strong Password")
else:
    print("Invalid Password")
```

IDLE Shell 3.13.9

File Edit Shell Debug Options Window Help

```
Python 3.13.9 (tags/v3.13.9:8183fa5, Oct 14 2025, 14:01:00) [AMD64] on win32
Enter "help" below or click "Help" above for more info:
>>>
=== RESTART: C:/Users/Pooja Sri/AppData/Local/Programs/Python/Python313/IDLE Shell 3.13.9
Enter Email ID: pooja26@gmail.com
Enter Password: Pooja@1307
Valid Email ID
Strong Password
>>> |
```

Section 5: Compile Once, Validate Many

A company receives 1,000 email IDs from its customers and needs to validate each one efficiently. Instead of compiling the same regular expression for every email, write a Python program using the **re.compile()** method to compile the pattern once and reuse it to validate all the email IDs.

Problem Statement

A company receives a large number of customer email IDs and needs to validate them efficiently. Instead of creating the same regular expression repeatedly for each email, write a Python program using the **re.compile()** method to compile the regular expression once and reuse it to validate all the email IDs.

untitled

File Edit Format Run Options Window Help

```
import re
email_pattern = re.compile(
    r'^[A-Za-z0-9]+[A-Za-z0-9._%+-]*@[A-Za-z0-9.-]+\.(com|org|edu|in)$'
)
emails = [
    "pooja@gmail.com",
    "student123@yahoo.org",
    "user_name@college.edu",
    "abc+12@test.in",
    "invalid@email",
    "@gmail.com",
    "user@domain.xyz"
]
for email in emails:
    if email_pattern.fullmatch(email):
        print(email, "-> Valid Email")
    else:
        print(email, "-> Invalid Email")
```

=== RESTART: C:/Users/Pooja Sri/AppData/Local/Programs/Python/Python313/5.py ===

```
pooja@gmail.com -> Valid Email
student123@yahoo.org -> Valid Email
user_name@college.edu -> Valid Email
abc+12@test.in -> Valid Email
invalid@email -> Invalid Email
@gmail.com -> Invalid Email
user@domain.xyz -> Invalid Email
```

Rubrics for Calculation

Criteria	Weight age	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Q. No. / Criteria	Conceptual Understanding	Accuracy of Answer	Application of Knowledge	Completeness of Response	Clarity & Presentation	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____